

Lymphocyte count and risk of chronic lymphocytic leukemia

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Abstract

Background: Elevated absolute lymphocyte count is often used as a first indication for diagnostic investigation of chronic lymphocytic leukemia (CLL). The absolute risk of CLL as a function of age, sex, and absolute lymphocyte count has not been described.

Methods: We used information about prediagnostic absolute lymphocyte cell count on 475 399 longitudinally followed research participants from UK Biobank, with a median follow-up of 11.6 years, during which 854 participants were diagnosed with CLL. A CLL risk model was developed based on sex, age at recruitment, and prediagnostic absolute lymphocyte count using flexible parametric survival modelling.

Results: Compared with research participants who had absolute lymphocyte counts below $2 \times 10^9/L$, the risk of CLL was almost 8 times higher for individuals with absolute lymphocyte counts of 3 to $4 \times 10^9/L$ (hazard ratio = 7.76, 95% CI = 6.24 to 9.65). The absolute risk of CLL was 2-fold higher in male than in female individuals. For 60-year-old persons with absolute lymphocyte counts of $5 \times 10^9/L$, 10-year CLL risk was estimated at 6.5% (95% CI = 5.6% to 7.5%) for men and 3.5% (95% CI = 3.0% to 4.1%) for women. The 10-year risk of CLL for participants with absolute lymphocyte counts of $5 \times 10^9/L$ varied between 0.91% (95% CI = 0.68% to 1.2%) for a 40-year-old woman and 12% (95% CI = 11% to 15%) for a 70-year-old man.

Conclusion: The absolute risk of CLL is strongly influenced by age, sex, and absolute lymphocyte count. A crude absolute lymphocyte count cutoff to indicate further investigation for CLL without taking age and sex into consideration will identify some individuals who are at low risk. These results may be considered when establishing guidelines for clinical management in patients with elevated absolute lymphocyte counts.

Introduction

Chronic lymphocytic leukemia (CLL) is characterized by mature B cells with a monoclonal population greater than $5 \times 10^9/L$ B lymphocytes in peripheral blood.^{1–3} The best established risk factors for CLL are age, sex, race, and common inherited genetic variants; relatives of patients with CLL have approximately 5- to 7-fold increased risk of CLL.^{4–6} Although CLL remains indolent in many patients⁷ and often does not require treatment at diagnosis, its diagnosis is associated with adverse health outcomes, including cytopenias, transformation to aggressive lymphomas, an increased risk of developing solid tumors, and immune dysfunction leading to impaired vaccine responses and heightened infection risk.^{8,9}

An elevated lymphocyte count, referred to as *lymphocytosis*, is often used as a first indication for further diagnostic investigation. Chronic lymphocytic leukemia is the most common diagnosis for patients with a hematological malignancy following investigations for lymphocytosis.⁸ The majority of patients with CLL—70% to 80%—are diagnosed with asymptomatic early-stage

disease through incidental detection of elevated lymphocyte counts.¹⁰ The relative and absolute risks of CLL as a function of age, sex, and lymphocyte count, however, has not previously been described.

We aimed to quantify the relative and absolute risk of CLL as a function of lymphocyte count, age, and sex based on prediagnostic data from nearly 500 000 participants in the UK Biobank.

Methods

Study population

The UK Biobank recruited more than 500 000 individuals aged 40 to 69 years between 2006 and 2010.¹¹ Participants were invited to the closest UK Biobank assessment center, where they provided written informed consent, completed study questionnaires and physical measurement, and donated baseline biospecimens (blood, urine, and saliva).¹² Absolute lymphocyte counts were measured on baseline whole blood centrally at the UK Biobank

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laboratory on all participants within 24 hours of blood draw using the Beckman Coulter LH750 analyzer.¹³

Study participants with CLL were defined based on International Statistical Classification of Diseases, Tenth Revision (ICD-10) codes C911 and C830 from the UK National Cancer registries data that are linked to UK Biobank.¹¹ Small lymphocytic lymphoma (ICD-10 code C830) is a lymphoma with similar cell phenotype to CLL but with fewer than $5 \times 10^9/L$ clonal B lymphocytes in the peripheral blood or the presence of lymphadenopathy. Small lymphocytic lymphoma is considered to belong to the same disease entity as CLL,¹⁴ and we hereafter refer to CLL or small lymphocytic lymphoma jointly as CLL.

We excluded individuals with a diagnosis of any prevalent hematological malignancy (ICD-10 code C81-96 or D45-D47) before recruitment to UK Biobank as well as individuals without a blood sample collected at baseline (missing at random), leaving 475 399 UK Biobank participants. After a median follow-up of 11.6 years (IQR: 10.9-12.6), 854 incident cases of CLL were diagnosed.

The study was approved by the Ethical Review Board of the International Agency for Research on Cancer, approval No. IEC 21-23.

Statistical analysis

The association between absolute lymphocyte count and CLL risk was initially assessed using Cox proportional hazards regression, with adjustment for age (continuous, in years) and sex (no missing data), with hazard ratios estimated for predefined categories of lymphocyte counts. Follow-up time (time from recruitment to diagnosis, death, or end of follow-up, whichever came first) was used as the time scale. The proportional hazard assumption was not met due to the distribution of lymphocyte count in the UK Biobank population, so we used flexible parametric survival models to address this situation, instead. Flexible parametric survival models with the same time-scale models were used to assess the absolute risk of being diagnosed with CLL within 10 years. Absolute lymphocyte count was modelled with a $\log_2$ transformation and polynomial cubic splines with 3 internal knots (located at 4, 8, and $10 \times 10^9/L$ B cells). The flexible parametric survival model parameters and the model equation can be found in the [Supplementary Material, Supplementary Methods](#). The performance model was assessed internally using 10-fold cross-validation by randomly sampling 70% of the UK Biobank study population for model training and using the remaining 30% as a testing set. Absolute risks were projected for

risk factor contrasts defined by age, sex, and lymphocyte count using a model that was trained on the entire UK Biobank dataset. As a sensitivity analysis, we refitted the risk model and re-estimated absolute risks after excluding participants with absolute lymphocyte counts above $10 \times 10^9/L$.

Results

Characteristics of participants

The final study sample included 475 399 participants, with a baseline median absolute lymphocyte count of $1.9 \times 10^9/L$ (IQR = 1.5 - $2.3 \times 10^9/L$). The median lymphocyte count at baseline for the 854 participants subsequently diagnosed with CLL was $2.77 \times 10^9/L$ (IQR = 2.0 - $5.1 \times 10^9/L$) (Table 1). Among participants diagnosed with CLL during follow-up, 218 had absolute lymphocyte counts above $5 \times 10^9/L$ at baseline, and 102 had counts above $10 \times 10^9/L$, with these elevated lymphocyte counts at baseline for some individuals preceding their CLL diagnosis more than 10 years. Among participants with no CLL diagnosis during follow-up, 502 (0.10%) had lymphocyte counts above $5 \times 10^9/L$, of whom 90 (0.02%) had lymphocyte counts above $10 \times 10^9/L$.

The association between lymphocyte count and CLL risk

We observed a strong dose-response relationship between absolute lymphocyte count and CLL risk, both for lymphocyte counts typically considered to be in the normal range (ie, $<4 \times 10^9/L$) and for counts above the normal range ($>4 \times 10^9/L$). Figure 1 shows the relative risk of incident CLL compared with individuals with absolute lymphocyte counts below $2 \times 10^9/L$. The CLL hazard ratio was 7.76 (95% CI = 6.24 to 9.65) for individuals with absolute lymphocyte counts of 3 to $4 \times 10^9/L$, 136 (95% CI = 94.2 to 197) for individuals with counts of 5 to $6 \times 10^9/L$, and 551 (95% CI = 356 to 855) for individuals with counts of 7 to $8 \times 10^9/L$. Above absolute lymphocyte counts of $8 \times 10^9/L$, the dose response relationship between lymphocyte count and CLL risk appeared to plateau.

Modelling the absolute risk of CLL based on age, sex, and absolute lymphocyte count

We subsequently developed a risk model using information about sex, age, and absolute lymphocyte count to estimate 10-year risks of being diagnosed with CLL. The CLL risk model displayed excellent internal performance in terms of model discrimination and calibration, suggesting that the model accurately reflects the risk of CLL the study participants experienced.

Table 1. Baseline characteristics of UK Biobank participants, by participants' CLL/SLL diagnosis status.

Characteristic	Participants diagnosed with CLL/SLL (n = 854)	Participants not diagnosed with CLL/SLL (n = 474 545)
Age at blood draw, median (IQR), y	63 (59-66)	58 (51-64)
Female sex, No. (%)	333 (39.0)	257 261 (54.2)
Lymphocyte count, median (IQR), $\times 10^9/L$	2.77 (1.98-5.09)	1.88 (1.51-2.30)
Lymphocyte category, $\times 10^9/L$	No. (%)	No. (%)
0-2	232 (27.2)	286 050 (60.3)
>2-3	228 (26.7)	163 433 (34.4)
>3-4	126 (14.8)	22 356 (4.7)
>4-5	50 (5.9)	2204 (0.5)
>5-6	32 (3.7)	274 (0.06)
>6-7	24 (2.8)	70 (0.01)
>7-8	22 (2.6)	30 (0.006)
>8-9	18 (2.1)	24 (0.005)
>9-10	20 (2.3)	14 (0.003)
>10	102 (11.9)	90 (0.02)

Abbreviations: CLL = chronic lymphocytic leukemia; SLL = small lymphocytic lymphoma.

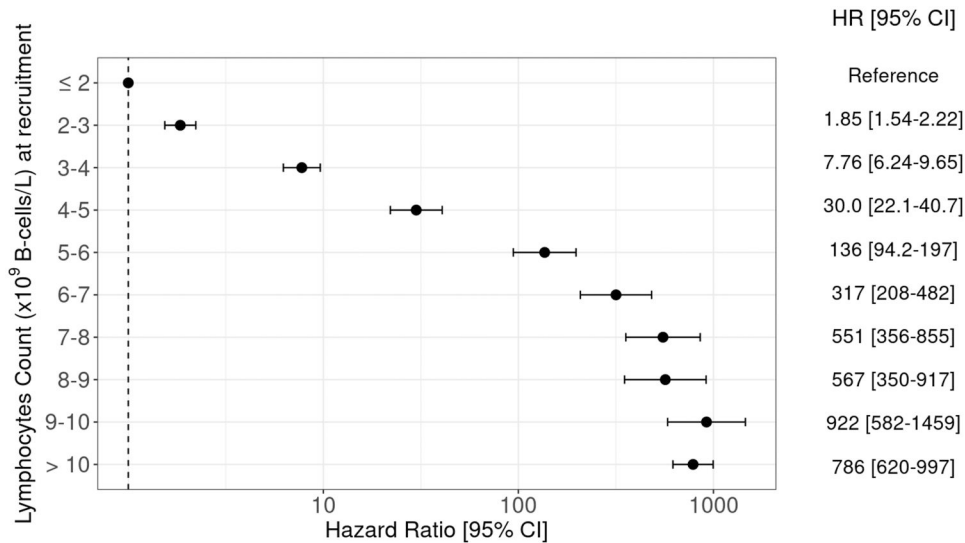


Figure 1. Relative risks with 95% CIs of chronic lymphocytic leukemia from our multivariable Cox regression model, including age (years), sex, and lymphocyte counts.

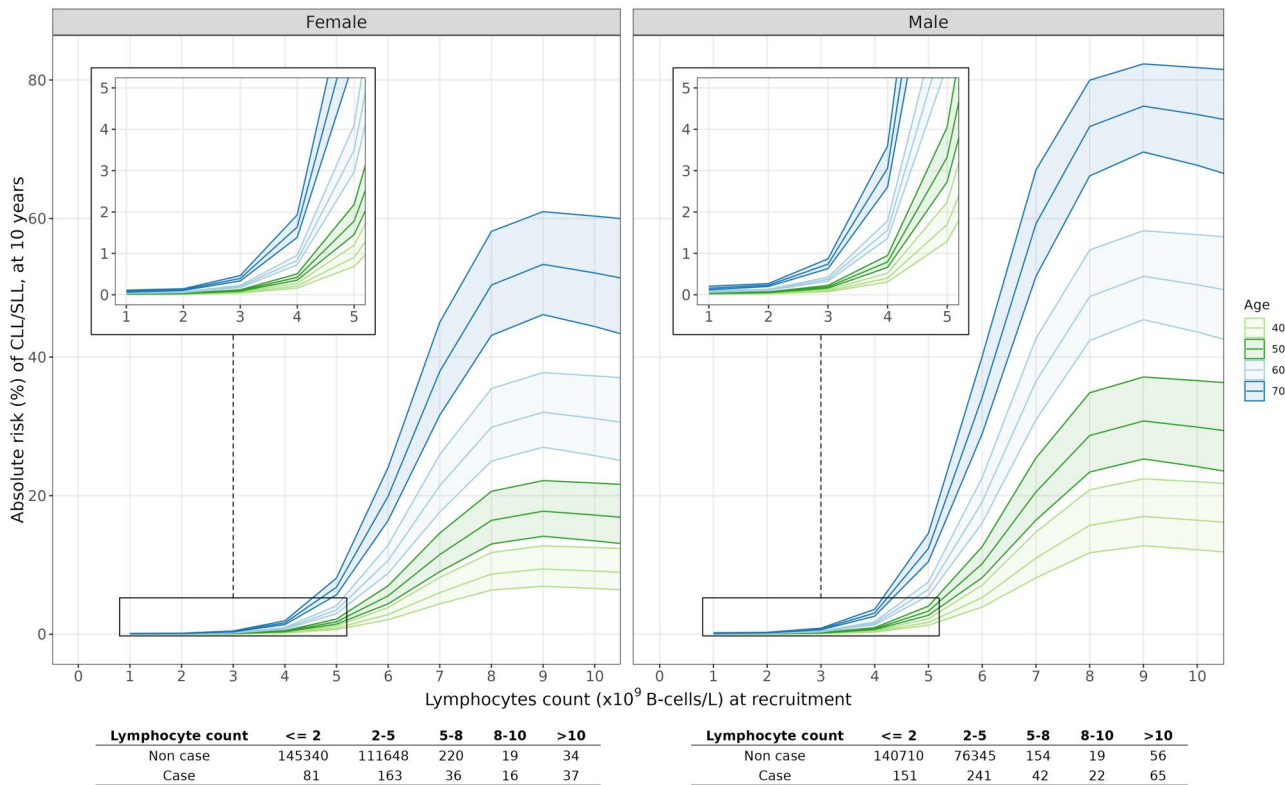


Figure 2. Estimated 10-year risk of being diagnosed with CLL based on age, lymphocyte count, and sex. Estimates are shown as a function of lymphocyte count, for women (left panel) and men (right panel) and stratified by age at recruitment. Histograms show the distribution of UK Biobank participants by lymphocyte counts. Abbreviations: CLL = chronic lymphocytic leukemia; SLL = small lymphocytic lymphoma.

Across the 10-fold cross-validation, the average area under curve (AUC) at 10years of follow-up was estimated at 0.84 (95% CI=0.82 to 0.86). The model calibration, the ratio between the expected and observed number of CLL cases, was estimated at 1.09 (95% CI=0.96 to 1.26). Calibration and discrimination across lymphocyte strata and risk percentile are provided in the [Supplementary Material, Figure S1](#).

Using a CLL risk model trained on the entire UK Biobank study population, we depicted the 10-year risk of being diagnosed with

CLL as a function of age, sex, and absolute lymphocyte count in [Figure 2](#). The CLL risk largely increases by age, and men have a risk almost double (hazard ratio=1.88, 95% CI=1.64 to 2.16) that of women with the same absolute lymphocyte count. [Tables 2](#) and [3](#) provide absolute CLL risk estimates by categories of absolute lymphocyte count, age, and sex. As an example, we estimated that a 50-year-old man with an absolute lymphocyte count of $5 \times 10^9/L$ has a 10-year CLL risk of 3.33% (95% CI = 2.73% to 4.05%) compared with 0.06% (95% CI=0.05%-0.07%) for a

Table 2. Absolute risk (%) of being diagnosed with chronic lymphocytic leukemia within 10 years for men, by lymphocyte count and exact age.

Lymphocyte count, $\times 10^9/L$	40 y, absolute risk % (95% CI)	45 y, absolute risk % (95% CI)	50 y, absolute risk % (95% CI)	55 y, absolute risk % (95% CI)	60 y, absolute risk % (95% CI)	65 y, absolute risk % (95% CI)	70 y, absolute risk % (95% CI)
1	0.02 (0.01-0.03)	0.03 (0.02-0.04)	0.04 (0.03-0.05)	0.06 (0.04-0.07)	0.08 (0.06-0.10)	0.11 (0.09-0.14)	0.16 (0.12-0.21)
2	0.03 (0.02-0.04)	0.04 (0.03-0.05)	0.06 (0.05-0.07)	0.08 (0.07-0.10)	0.12 (0.11-0.13)	0.17 (0.15-0.19)	0.23 (0.20-0.27)
3	0.10 (0.07-0.12)	0.14 (0.11-0.17)	0.19 (0.16-0.23)	0.27 (0.23-0.31)	0.38 (0.33-0.43)	0.53 (0.46-0.60)	0.74 (0.63-0.87)
4	0.40 (0.31-0.52)	0.56 (0.45-0.70)	0.79 (0.66-0.95)	1.11 (0.96-1.29)	1.56 (1.36-1.79)	2.19 (1.90-2.51)	3.06 (2.60-3.59)
5	1.70 (1.29-2.24)	2.38 (1.88-3.01)	3.33 (2.73-4.05)	4.65 (3.92-5.50)	6.47 (5.55-7.53)	8.97 (7.70-10.4)	12.4 (10.5-14.6)
6	5.28 (3.92-7.09)	7.34 (5.68-9.46)	10.2 (8.15-12.6)	14.0 (11.5-16.9)	19.1 (16.1-22.6)	25.7 (21.8-30.2)	34.2 (28.9-40.1)
7	11.0 (8.13-14.8)	15.1 (11.6-19.4)	20.5 (16.5-25.4)	27.6 (22.9-33.1)	36.5 (31.0-42.7)	47.2 (40.7-54.2)	59.2 (51.6-67.0)
8	15.7 (11.8-20.8)	21.4 (16.7-27.1)	28.7 (23.4-34.8)	37.8 (32.0-44.3)	48.7 (42.4-55.5)	60.9 (54.1-67.8)	73.3 (66.1-80.0)
9	17.0 (12.8-22.4)	23.0 (18.1-29.0)	30.8 (25.3-37.1)	40.4 (34.4-46.9)	51.7 (45.4-58.2)	64.0 (57.5-70.5)	76.2 (69.6-82.3)
≥ 10	16.5 (12.2-22.0)	22.3 (17.3-28.5)	29.9 (24.2-36.6)	39.3 (33.0-46.4)	50.4 (43.6-57.7)	62.7 (55.5-69.9)	75.0 (67.7-81.8)

Table 3. Absolute risk (%) of being diagnosed with chronic lymphocytic leukemia within 10 years for women, by lymphocyte count and age.

Lymphocyte count, $\times 10^9/L$	40 y, absolute risk % (95% CI)	45 y, absolute risk % (95% CI)	50 y, absolute risk % (95% CI)	55 y, absolute risk % (95% CI)	60 y, absolute risk % (95% CI)	65 y, absolute risk % (95% CI)	70 y, absolute risk % (IQR95% CI)
1	0.01 (0.01-0.02)	0.01 (0.01-0.02)	0.02 (0.02-0.03)	0.03 (0.02-0.04)	0.04 (0.03-0.06)	0.06 (0.05-0.08)	0.08 (0.06-0.11)
2	0.02 (0.01-0.02)	0.02 (0.02-0.03)	0.03 (0.03-0.04)	0.04 (0.04-0.05)	0.06 (0.06-0.07)	0.09 (0.08-0.10)	0.12 (0.11-0.15)
3	0.05 (0.04-0.07)	0.07 (0.06-0.09)	0.10 (0.09-0.12)	0.14 (0.12-0.16)	0.20 (0.18-0.23)	0.28 (0.24-0.32)	0.39 (0.33-0.46)
4	0.21 (0.16-0.28)	0.30 (0.24-0.37)	0.42 (0.35-0.51)	0.59 (0.51-0.69)	0.83 (0.72-0.96)	1.17 (1.01-1.35)	1.64 (1.39-1.93)
5	0.91 (0.68-1.20)	1.27 (1.00-1.61)	1.78 (1.45-2.18)	2.50 (2.09-2.98)	3.49 (2.97-4.10)	4.87 (4.14-5.73)	6.78 (5.67-8.10)
6	2.84 (2.09-3.85)	3.97 (3.04-5.18)	5.54 (4.38-6.98)	7.69 (6.25-9.45)	10.6 (8.78-12.9)	14.6 (12.1-17.6)	19.9 (16.4-24.1)
7	5.99 (4.37-8.20)	8.32 (6.31-10.9)	11.5 (9.02-14.6)	15.8 (12.7-19.5)	21.4 (17.6-25.9)	28.8 (23.9-34.4)	37.9 (31.6-45.0)
8	8.69 (6.37-11.8)	12.0 (9.16-15.6)	16.4 (13.0-20.6)	22.3 (18.2-27.1)	29.9 (25.0-35.5)	39.3 (33.3-45.9)	50.4 (43.1-58.1)
9	9.42 (6.92-12.8)	13.0 (9.95-16.8)	17.8 (14.1-22.2)	24.0 (19.8-29.0)	32.0 (27.0-37.8)	41.9 (35.9-48.5)	53.4 (46.1-61.0)
≥ 10	9.11 (6.60-12.5)	12.6 (9.49-16.5)	17.2 (13.5-21.8)	23.3 (18.8-28.6)	31.1 (25.8-37.3)	40.8 (34.4-47.9)	52.1 (44.4-60.3)

50-year-old man with an absolute lymphocyte count of $2 \times 10^9/L$ (Table 2). Corresponding 10-year risk estimates for a 50-year-old woman were 1.78% (95% CI=1.45% to 2.18%) and 0.03% (95% CI=0.03% to 0.04%), respectively (Table 3). For individuals with absolute lymphocyte counts of $5 \times 10^9/L$, the 10-year risk of CLL varied between 0.91% (95% CI=0.68% to 1.20%) for a 40-year-old woman (Table 3) and 12.4% (95% CI=10.5% to 14.6%) for a 70-year-old man (Table 2). In a sensitivity analysis, we excluded participants with absolute lymphocyte counts above $10 \times 10^9/L$ and refitted the risk model, which yielded similar performance as the main model (expected/observed=1.06, 95% CI=0.95 to 1.22; AUC=0.84, 95% CI=0.83 to 0.87). Risk estimates based on this model are provided in the [Supplementary Material, Tables S1 and S2](#).

Discussion

Our study provides the first comprehensive description of how lymphocyte count, age, and sex affect the risk of subsequent CLL diagnosis. We found that a single measurement of absolute lymphocyte count is a robust risk indicator of subsequent CLL diagnosis for individuals with counts both within and above the normal ($\leq 4 \times 10^9/L$) absolute lymphocyte range. We provide comprehensive 10-year CLL risk estimates as a function of age, sex, and absolute lymphocyte count.

Elevated absolute lymphocyte count is often used in medical practice as the first indication for further diagnostic investigation of CLL. Randomized clinical trials of early treatment of asymptomatic patients with CLL have thus far not led to improved overall survival, and treatment is not recommended for asymptomatic patients diagnosed with CLL.^{2,3,15} The benefit for the individual

of an early CLL diagnosis is therefore a matter of debate,¹⁶⁻¹⁸ and there is no consensus as to when elevated lymphocyte levels should prompt further investigation. Many general practice guidelines use an absolute lymphocyte count above $5 \times 10^9/L$ as a threshold for referral to a hematologist, but both lower and higher thresholds are commonly used.^{7,19,20} These guidelines, however, do not consider the differences in absolute risk of CLL for a given absolute lymphocyte count experienced by individuals of different ages and sexes.

We found that prediagnostic absolute lymphocyte count was strongly associated with risk of subsequent CLL diagnosis, including for counts within the range considered normal (ie, $\leq 4 \times 10^9/L$). Compared with participants with absolute lymphocyte counts below $2 \times 10^9/L$, we observed a notable log-linear trend in increasing CLL risk up to counts of $8 \times 10^9/L$. The increasing risk is likely partially attributable to a higher prevalence of high count monoclonal B lymphocytosis in individuals with elevated lymphocyte counts, reflecting the known association between high count monoclonal B lymphocytosis and progression to CLL.^{7,21} Monoclonal B lymphocytosis is defined as the presence of a clonal B lymphocyte population less than $5 \times 10^9/L$ B lymphocytes and no disease-related symptoms.^{7,22} Monoclonal B lymphocytosis has been observed in up to 12% of individuals older than 40 years of age and in 20% of siblings of patients with CLL; it can be present 15 to 25 years before diagnosis of CLL.^{7,22,23} Although most individuals with monoclonal B lymphocytosis will never develop CLL, those with a B lymphocyte clone of 0.5 to $5 \times 10^9/L$, known as high-count monoclonal B lymphocytosis, have a 1% to 2% yearly risk of progression to CLL that requires treatment.^{7,21}

To assess the absolute risk of CLL, we developed a risk model based on age, sex, and lymphocyte count that accurately

estimated the 10-year absolute risk of CLL in the UK Biobank dataset. Using this risk model, we found that the absolute risk of CLL at any given lymphocyte count depends strongly on age and sex (Figure 1). For example, we estimated that the 10-year CLL risk for participants with a lymphocyte count of $5 \times 10^9/L$ ranged from 1% for women aged 40 years to 13% for men aged 70 years (Figure 1 and Table 2). To contextualize this level of risk, a 5-year risk of 5% (ie, equivalent of >10% 10-year risk) is commonly used to indicate surveillance for solid cancers, such as breast, cervical, and lung cancers.²⁴ Although early detection in those cancers improves outcomes through curative therapies, this is not currently the case for CLL. Our findings show that many individuals above commonly used clinical thresholds (eg, absolute lymphocyte count $>5 \times 10^9/L$) remain at relatively low risk of being subsequently diagnosed with CLL.^{2,3,15} This finding highlights a need for research into identifying absolute risk thresholds—or sex-specific and age-specific lymphocyte count thresholds—for investigation of CLL that take the potential benefits and harms for the individual into consideration, including the cost-effectiveness and efficacy of available treatments.

An additional consideration is the radically changed treatment landscape for CLL in recent decades. Several oral agents have been approved, including Bruton tyrosine kinase inhibitors and venetoclax, and next-generation anti-CD20 monoclonal antibodies, such as obinutuzumab. There are also novel and improved tools for prediction of disease progression.²⁵ Ongoing trials continue to explore the potential benefits of novel therapies in asymptomatic patients at high risk of progression.^{26,27} If these studies show clear benefits (and minimal harms) of early interventions, it could shift the cost-benefit balance in favor of diagnosing asymptomatic patients earlier. Similarly, if additional markers, such as genetic factors,^{28,29} or other risk-stratification tools^{22,30} could better identify individuals at risk of developing symptomatic disease, particularly in individuals with absolute lymphocyte counts between 4 and $5 \times 10^9/L$ (Tables 2 and 3), this could shift the risk-benefit balance in favor of earlier recognition of CLL and mitigation of related adverse health outcomes.

Our findings support reconsidering current guidelines for evaluating lymphocyte counts and to define age-specific and sex-specific thresholds for further investigation to ensure that the benefits of referral to hematology or to surveillance outweigh potential harms.

Study strengths and limitations

Our study has several strengths and limitations that should be considered when interpreting the findings.

First, in contrast to a clinical setting, we had access to only a single blood lymphocyte count per study participant. We were therefore unable to evaluate the predictive value of either intra-individual repeated lymphocyte counts for estimating CLL risk or the presence of a monoclonal cell population in the blood sample. Second, because of the nature of the collected data, we had limited information to examine why certain individuals had elevated lymphocyte counts that in the clinical setting would normally be considered, such as infections and inflammatory disease. Most importantly, the lymphocyte counts used to train the CLL risk model study were not considered in the diagnostic pipeline of the CLL cases in our study. This means that absolute risk estimates generated by our CLL model are not directly influenced—or biased—by the lymphocyte counts used to generate the CLL prediction model. Although we consider this an important strength of our study, elevated lymphocyte counts taken in a clinical setting would likely have higher positive predictive

values than those estimated here (ie, higher absolute risks) because they might trigger subsequent diagnostic workup, thereby increasing the likelihood (ie, risk) of subsequent CLL diagnosis. We also note that, considering the often indolent nature of CLL,²¹ it is likely that some participants in UK Biobank have undiagnosed CLL. In a sensitivity analysis, we therefore excluded participants with lymphocyte counts above $10 \times 10^9/L$ and found similar risk estimates (Supplementary Material, Table S2).

A major strength of our study is the availability of a large population cohort with close to 500 000 participants with baseline lymphocyte counts that were longitudinally followed. This population enabled us to develop a robust risk model for CLL that was internally validated. We are not aware of similar study resources that could be used to externally validate our CLL risk model. This means that we have no external evidence to justify translating our risk model into clinical practice. Considering the accuracy by which our risk model estimated CLL risk in UK Biobank, however, the risk estimates reported in Table 2 should be considered when developing new guidelines for workup of patients with elevated lymphocyte counts along with any emerging evidence of benefit from early interventions following CLL diagnosis.

Conclusions

This study demonstrates that the absolute risk of CLL is strongly influenced by age, sex, and absolute lymphocyte count. We found that using a crude lymphocyte count cutoff to indicate further investigation for CLL without taking age and sex into consideration will identify some individuals who are at low risk. This study provides 10-year risk estimates that may be considered when establishing guidelines for clinical management of patients with elevated lymphocyte counts.

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Author contributions

Simon Pahnke (Conceptualization; Data curation; Formal analysis; Writing—original draft; Writing—review & editing), Karine Alcalá (Data curation; Formal analysis; Methodology; Writing—original draft; Writing—review & editing), Connie Bulos Salim (Validation; Writing—review & editing), Ricardo Cortez Cardoso Penha (Writing—review & editing), HanLa Park (Writing—review & editing), James McKay (Conceptualization; Funding acquisition; Resources; Supervision; Writing—original draft; Writing—review & editing), and Mattias Johansson (Conceptualization; Funding acquisition; Methodology; Resources; Supervision; Writing—original draft; Writing—review & editing)

Supplementary material

Supplementary material is available at JNCI: Journal of the National Cancer Institute online.

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Conflicts of interest

The authors declare no conflicts of interest.

Where authors are identified as personnel of the International Agency for Research on Cancer/World Health Organization, the authors alone are responsible for the views expressed in this article, and they do not necessarily represent the decisions, policy, or views of the International Agency for Research on Cancer/World Health Organization.

Data availability

The observational data used for this study are available upon request to the UK Biobank.

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